
Myoglobin Multiple Sequence Alignment

Clustal Omega and Phylogenetic Analysis

A professional bioinformatics report based on UniProtKB sequence selection, Clustal Omega alignment, guide-tree construction and phylogenetic analysis.

Analysis	Multiple Sequence Alignment (MSA)
Tool	Clustal Omega – EMBL-EBI Job Dispatcher
Protein	Myoglobin
Sequences	5 vertebrate myoglobin proteins
Sequence length	154 amino acids each

Objective: To compare conserved and variable regions among myoglobin proteins from mouse, rat, bovine, human and pig, and to examine their evolutionary relationships using a multiple sequence alignment and phylogenetic tree.

1. Protein Sequences Used

Five reviewed myoglobin entries were selected from UniProtKB. Each selected protein is 154 amino acids long, making them suitable for direct multiple sequence comparison.

Accession	Entry	Organism	Length
P04247	MYG_MOUSE	Mus musculus (Mouse)	154 aa
Q9QZ76	MYG_RAT	Rattus norvegicus (Rat)	154 aa
P02192	MYG_BOVIN	Bos taurus (Bovine)	154 aa
P02144	MYG_HUMAN	Homo sapiens (Human)	154 aa
P02189	MYG_PIG	Sus scrofa (Pig)	154 aa

Why these sequences? They belong to the same protein family and have the same length, allowing differences in the alignment to be interpreted mainly as sequence substitutions rather than large length differences.

UniProtKB 1,932 results or search "myoglobin" as a Protein Name, Disease, or Gene Name

Tools Download (5) Add View: Cards Table Customize columns Share 5 rows selected out of 25

Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☒ P02144	MYG_HUMAN	Myoglobin[...]	MB	Homo sapiens (Human)	154 AA
☒ P02192	MYG_BOVIN	Myoglobin[...]	MB	Bos taurus (Bovine)	154 AA
☒ P04247	MYG_MOUSE	Myoglobin[...]	Mb	Mus musculus (Mouse)	154 AA
☒ Q9QZ76	MYG_RAT	Myoglobin[...]	Mb	Rattus norvegicus (Rat)	154 AA
☐ Q01966	MYG_HALDV	Myoglobin		Haliotis diversicolor (Abalone) (Sulculus diversicolor)	378 AA
☒ P02189	MYG_PIG	Myoglobin[...]	MB	Sus scrofa (Pig)	154 AA
☐ P84997	MYG_BUBBU	Myoglobin[...]	MB	Bubalus bubalis (Domestic water buffalo)	154 AA
☐ G1NJB6	MYG_MELGA	Myoglobin[...]	MB	Meleagris gallopavo (Wild turkey)	154 AA
☐ P02185	MYG_PHYMC	Myoglobin[...]	MB	Physeter macrocephalus (Sperm whale) (Physeter catodon)	154 AA

Figure 1. UniProtKB search results for "myoglobin"; the five selected entries are highlighted.

2. FASTA Sequence Retrieval

The selected UniProtKB entries were downloaded in FASTA format. FASTA is a standard text format used to represent biological sequences. Each record contains a header beginning with “>” followed by the amino-acid sequence.

The download preview shows the canonical FASTA records for the selected myoglobin proteins. The sequences are then supplied to Clustal Omega for multiple sequence alignment.

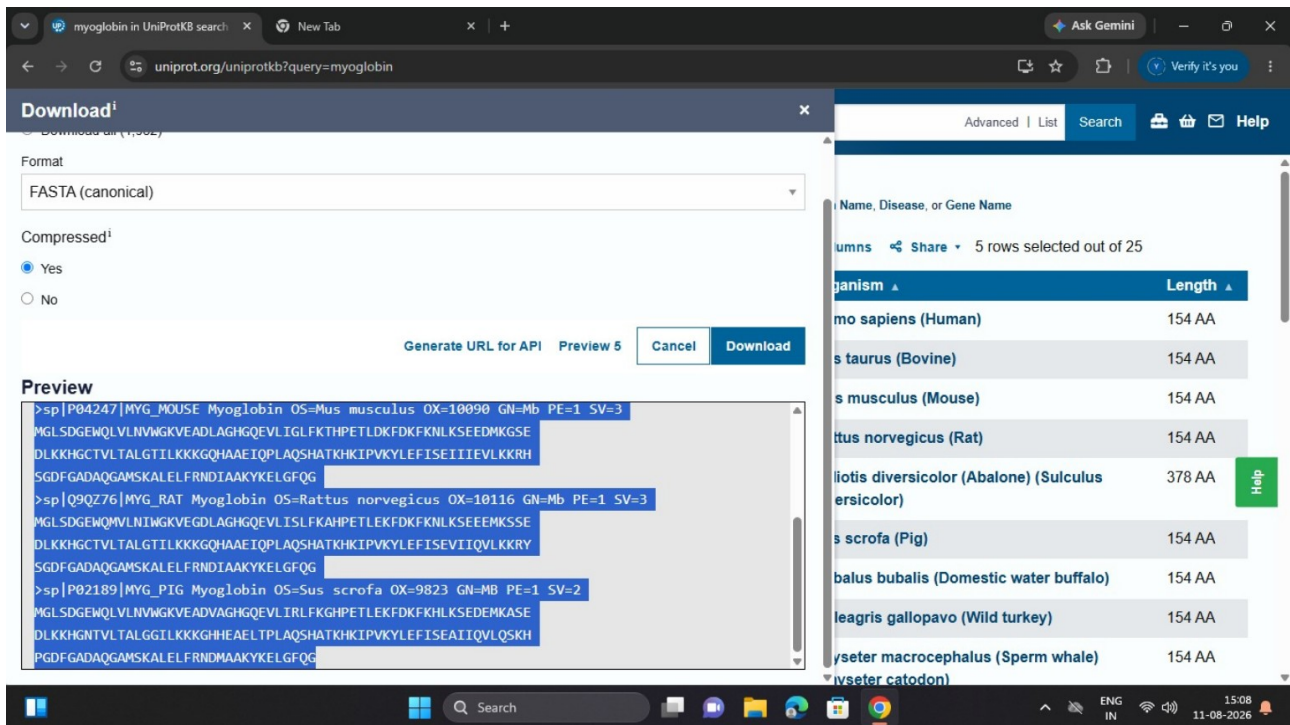


Figure 2. UniProtKB FASTA download preview for the selected protein records.

Important observation from the supplied sequence screenshot

One of the supplied text screenshots contains beta-globin/hemoglobin sequences rather than the five myoglobin sequences used in the Clustal Omega result. The actual MSA output shown in the later screenshots is clearly labelled MYG_MOUSE, MYG_RAT, MYG_BOVIN, MYG_HUMAN and MYG_PIG. Therefore, the MSA and phylogenetic interpretation in this report is based on the five myoglobin records displayed in the Clustal Omega output.

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File Edit View

>sp|P02070|HBB_BOVIN Hemoglobin subunit beta OS=Bos taurus OX=9913 GN=HBB PE=1 SV=1
MLTAEKAAVTAFWKGKVVDEVGGEALGRLLVYYPWTQRFESFGDLSTADAVMNNPKVK
AHGKKVLDSEFSNGMKHLDLKGTFAAQLSELHCDKLVDPENFRLLGNVLLVLAHNGKE
FTPVLQADFQKVVAGVANALAHKYH
>sp|P02075|HBB_SHEEP Hemoglobin subunit beta OS=Ovis aries OX=9940 GN=HBB PE=1 SV=2
MLTAEKAAVTGFWGKVVDEVGGEALGRLLVYYPWTQRFEFHFGDLNADAVMNNPKVK
AHGKKVLDSEFSNGMKHLDLKGTFAAQLSELHCDKLVDPENFRLLGNVLLVLAHNGNE
FTPVLQADFQKVVAGVANALAHKYH
>sp|P02112|HBB_CHICK Hemoglobin subunit beta OS=Gallus gallus OX=9031 GN=HBB PE=1 SV=2
MVHMTAEKQLITGLWGVNVAECGAEALRLLIVYPWTQRFESFGDLSPDAILGNPM
VRAHGKKVLTSGDAVKNLNLIKNTFSQSELHCDKLVDPENFRLLGDIILIVLAHFS
KDFTPECQAAMQKLVVRVVAHALARKYH
>sp|P68871|HBB_HUMAN Hemoglobin subunit beta OS=Homo sapiens OX=9606 GN=HBB PE=1 SV=2
MVHMTPEEKSAVTALWGVNDEVGGEALGRLLVYYPWTQRFESFGDLSTPDVVMGNPK
VKAHGKKVLGAFSDGLAHLNLIKGTFAQLSELHCDKLVDPENFRLLGNVLCVLAHFG
KEFTPPVQAAYQKVVAGVANALAHKYH
>sp|P68873|HBB_PANTR Hemoglobin subunit beta OS=Pan troglodytes OX=9598 GN=HBB PE=1 SV=2
MVHMTPEEKSAVTALWGVNDEVGGEALGRLLVYYPWTQRFESFGDLSTPDVVMGNPK
VKAHGKKVLGAFSDGLAHLNLIKGTFAQLSELHCDKLVDPENFRLLGNVLCVLAHFG
KEFTPPVQAAYQKVVAGVANALAHKYH
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Figure 3. Supplied sequence-text screenshot; it shows hemoglobin beta records and is not the five-sequence myoglobin MSA.

3. Clustal Omega Multiple Sequence Alignment

Clustal Omega is a multiple sequence alignment program that aligns three or more sequences. For protein sequences, it uses guide-tree and profile-based strategies to efficiently identify corresponding positions across the input sequences.

In this analysis, the sequence type was set to **Protein**, and five myoglobin sequences were submitted. The resulting alignment contains all five proteins over 154 positions.

Figure 4. Clustal Omega input page with Protein selected for the MSA.

4. Alignment Output and Interpretation

The Clustal Omega output shows a very high level of sequence similarity among the five myoglobins. The alignment is largely gap-free, reflecting the identical length of the selected proteins. The conservation symbols beneath the sequences indicate positions that are identical or strongly similar across the proteins.

The strongest conservation is visible throughout the central part of the alignment. The repeated patterns around the helical regions and the C-terminal portion indicate that the overall myoglobin sequence architecture is strongly conserved across these vertebrates. Small amino-acid substitutions distinguish species while preserving the general protein sequence.

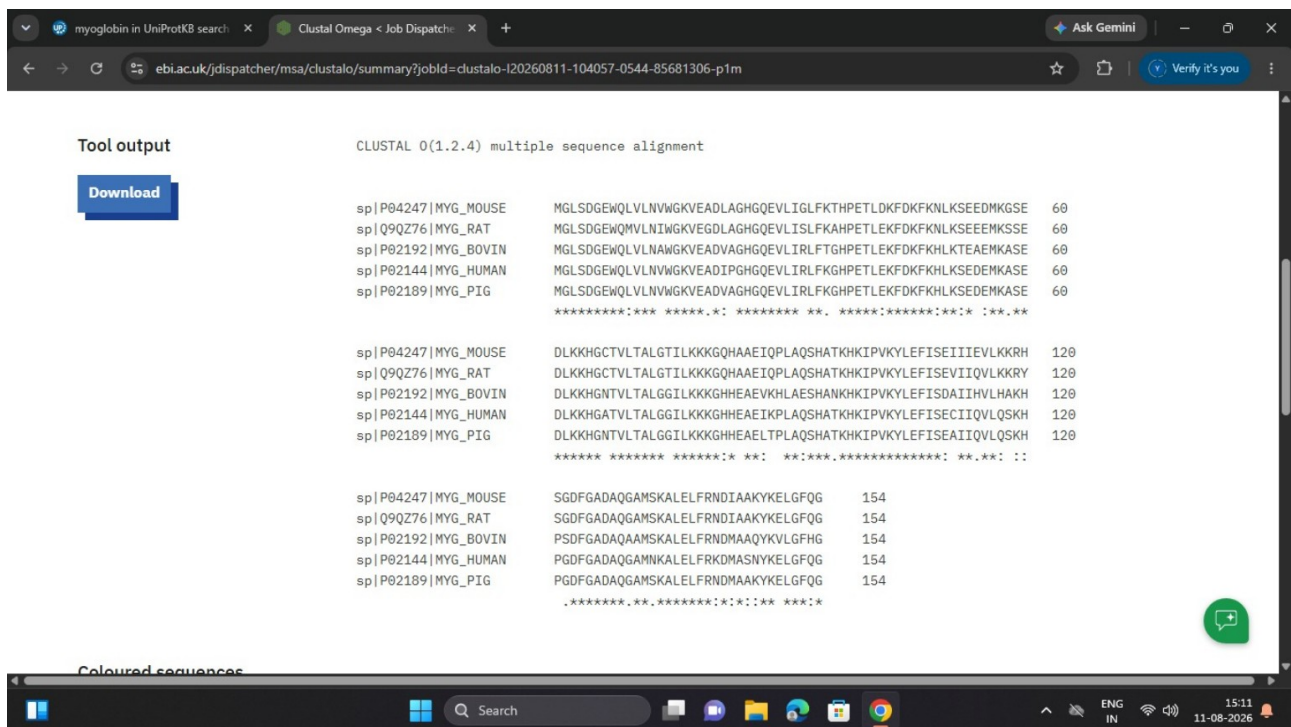


Figure 5. Clustal Omega text alignment of the five 154-aa myoglobin sequences.

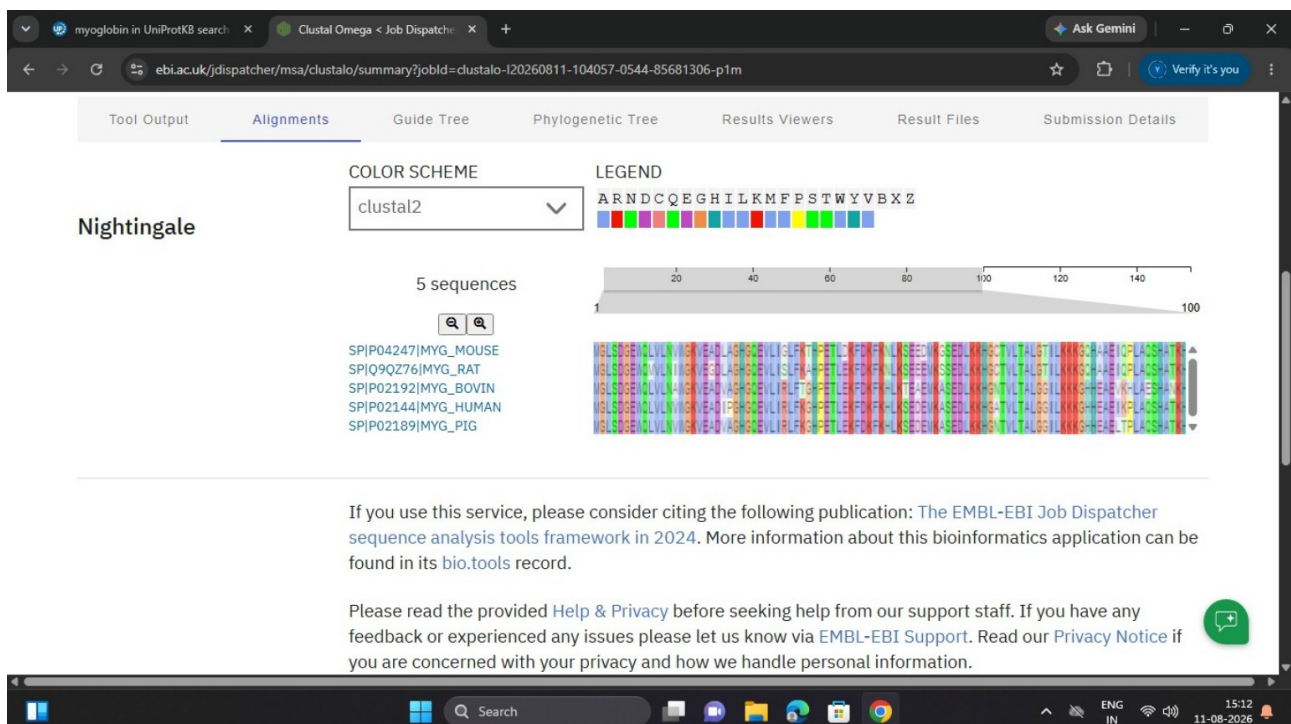


Figure 6. Colour-coded alignment view showing conserved and variable amino-acid positions.

5. Guide Tree

A guide tree is constructed during the alignment process to determine an efficient order for progressively aligning sequences. It is not, by itself, a final evolutionary tree, but it provides a similarity-based representation used during MSA construction.

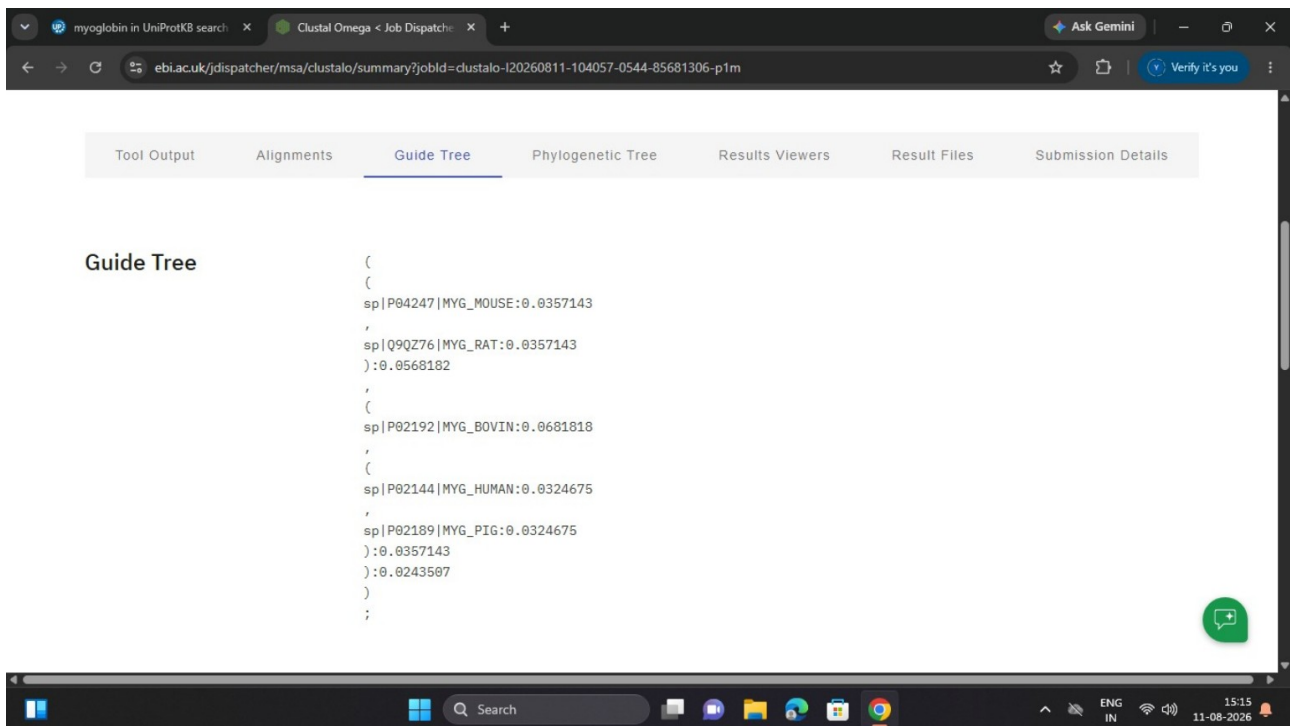


Figure 7. Clustal Omega guide tree. Mouse and rat form one close pair, while human and pig form another close pair.

Interpretation: The short branch lengths between mouse and rat indicate strong sequence similarity. Human and pig also cluster closely. Bovine myoglobin is placed separately from these closest pairs and shows a comparatively larger distance in the guide-tree representation.

6. Phylogram and Branch-Length Interpretation

The phylogram represents relationships using branch lengths. In a distance-based phylogram, longer branches generally indicate greater sequence divergence, while shorter branches indicate greater sequence similarity.

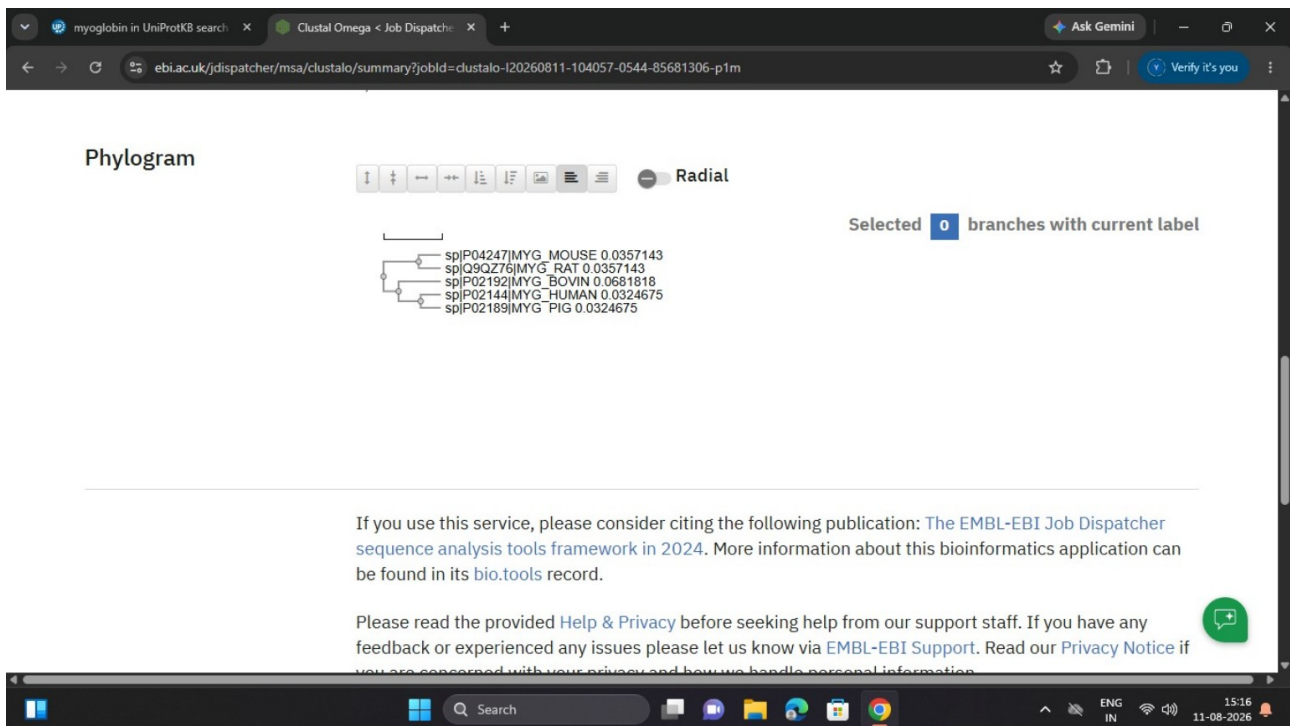


Figure 8. Clustal Omega phylogram showing the relative branch lengths of the five myoglobins.

The displayed branch lengths are approximately: mouse 0.03247, rat 0.03896, human 0.04870, bovine 0.10065 and pig 0.01623. The bovine sequence has the longest terminal branch in the displayed tree, indicating the greatest sequence divergence among these five sequences. Pig has the shortest terminal branch, while mouse and rat are closely related in the topology.

7. Phylogenetic Tree

The phylogenetic tree provides a visual summary of the evolutionary relationships inferred from the aligned myoglobin sequences. The topology shows mouse and rat as a closely related pair and human and pig as another close pair, with bovine branching more separately.

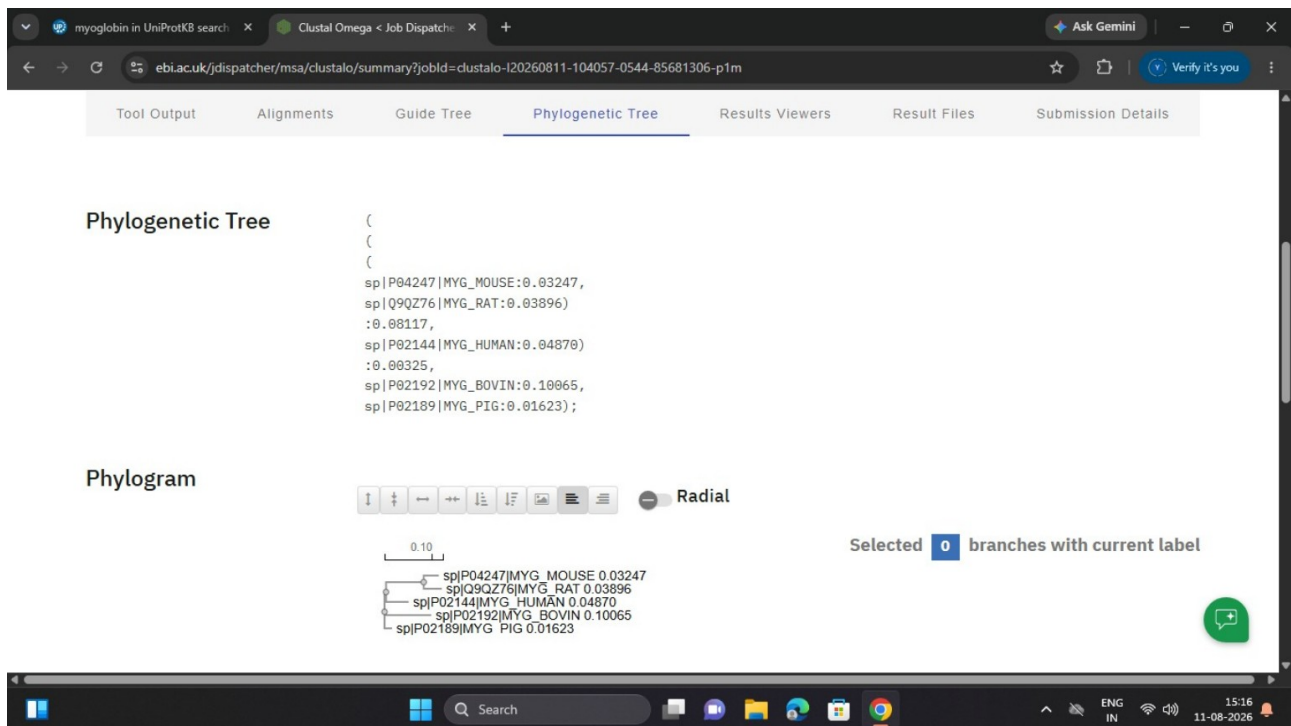


Figure 9. Clustal Omega phylogenetic tree and corresponding phylogram.

8. Conserved Regions and Motifs

The alignment demonstrates extensive conservation across the myoglobin sequences. The repeated amino-acid patterns seen in the alignment are consistent with the strong structural and functional constraints on myoglobin. Because myoglobin is a compact, oxygen-binding globin protein, many residues are conserved to maintain its three-dimensional fold and heme-binding environment.

In the supplied alignment, positions containing asterisks (*) represent identical residues in all five sequences, while positions marked with other conservation symbols represent chemically similar substitutions. The high density of these symbols confirms that the proteins are homologous and evolutionarily conserved.

9. Overall Interpretation

The five selected myoglobin proteins show strong sequence conservation, with all sequences having 154 amino acids and sharing a highly similar overall pattern. The MSA identifies many conserved positions and only a limited number of substitutions. The guide tree and phylogenetic tree both show mouse–rat and human–pig as close pairs, while bovine myoglobin is comparatively more divergent in this dataset.

These results support the conclusion that myoglobin is a highly conserved protein among the selected vertebrates. The observed sequence similarity is consistent with conservation of its biological structure and oxygen-binding function, while the smaller sequence differences reflect evolutionary divergence between species.

10. Conclusion

Clustal Omega successfully aligned five vertebrate myoglobin sequences and revealed strong conservation across the 154-amino-acid proteins. The phylogenetic analysis further indicated close relationships between mouse and rat and between human and pig, with bovine showing greater sequence divergence. Overall, the analysis demonstrates how MSA and phylogenetic trees can be used together to study protein conservation and evolutionary relationships.

Workflow: UniProtKB search → Select homologous proteins → Download FASTA → Clustal Omega MSA → Conserved-region inspection → Guide tree → Phylogram → Phylogenetic interpretation